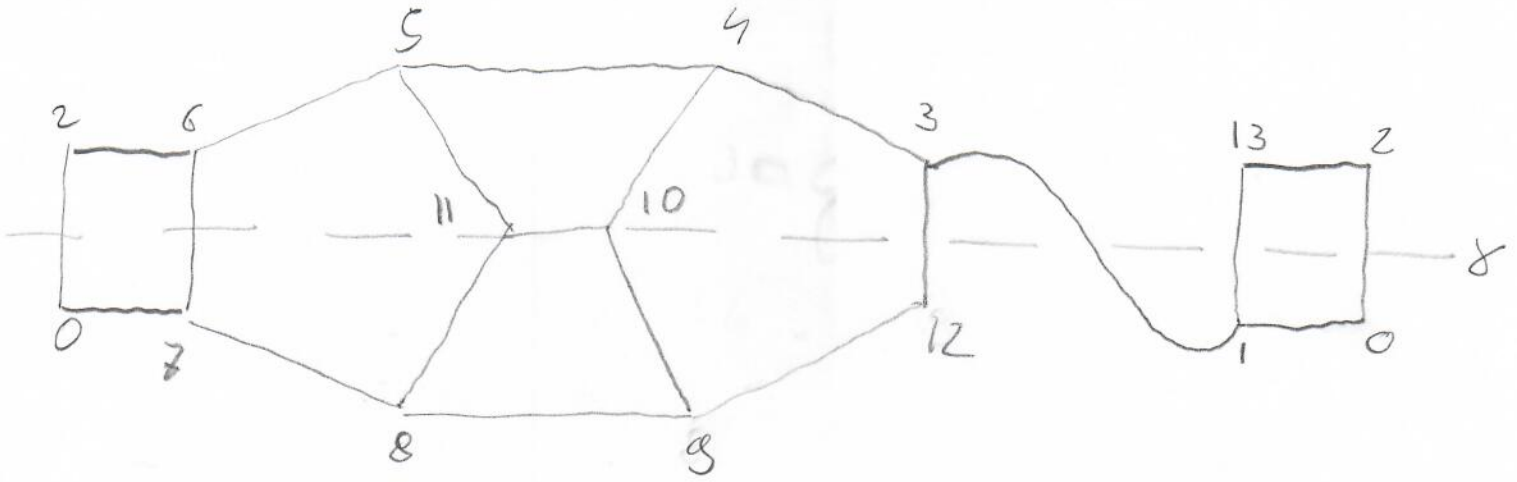
Net 12

We draw a diagram



Note that the pentagons are symmetric with respect to the equator γ containing the edge '11''10' which is also an axis of symmetry for the quadrilaterals '13'201 and 6207.

Consider the equator containing the edge 31. As we travel from 3 to 1 at the beginning the latitude of the point is increasing and it must reach its maximum before the equator crosses and connects to 1. A symmetric argument also implies that the edge 31 must contain the point of minimal latitude of γ

So the edge 31 must be longer than π
But the long edge lemma would imply that the face 2'13'134562 has area larger than $\frac{4\pi}{3}$, which is not possible